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Electroactive advanced nanomaterials for wastewater monitoring

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Abstract
The need for advanced water pollution monitoring and remediation methods has increased due to rapid development and human activity. This review emphasizes 'electroactive nanomaterials' importance in sustainable solutions, detailing their use in environmental water monitoring and treatment. The key materials discussed include: activated carbon, biochar-derived nanomaterials, graphene, graphene quantum dots, and metal-organic frameworks. These materials display outstanding electrochemical behaviour, tunable properties, and environmental adaptability. These nanomaterials show significant potential for real-time detection of various contaminants, making them ideal for future water monitoring. The discussion also covers potential future research, concentrating on scalable sensor technology, biocompatibility and material engineering for practical environmental applications.

Highlights

- Recent works (year 2018–2025) have been reviewed for wastewater monitoring through electrochemical method.
- MOFs, Graphene, GQDs, ACs and biochar-based nanomaterials have been explored.
- Real-time detection of harmful contaminant in wastewater.
- Future prospective focus of sustainable and cost-effective nanocomposite.

Keywords Electrochemical, Toxic pollutants, MOF, Graphene, GQDs, Activated charcoal, Biochar wastewater, Nanocomposite

1 Introduction

Environmental pollution is one of the most critical global challenges, significantly impacting human health and ecological balance. The uncontrolled discharge of waste from households, industrial processes, and agricultural activities has severely deteriorated the quality of water bodies, including rivers, lakes, and ponds. Aquatic ecosystems are increasingly contaminated with hazardous pollutants such as heavy metals, pesticides, and pharmaceutical residues, which pose severe health risks [1]. Pollution is defined as the presence of one or more undesirable chemical entities that disrupt natural processes or adversely affect living organisms and the environment. The rapid pace of industrialization and population growth, coupled with urban expansion, has accelerated

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the accumulation of environmental pollutants [2]. According to the World Health Organization, environmental pollution-including air, water, and soil contamination-is responsible for approximately 25% of global human diseases [3].

Addressing these multifaceted environmental problems requires the development of innovative scientific and technological strategies. Nanotechnology, which focuses on the design, synthesis, and manipulation of materials at the nanoscale, has emerged as a transformative approach in environmental remediation [4]. Nanomaterials exhibit unique physicochemical properties, such as high surface-to-volume ratios, enhanced reactivity, and tunable optical, electrical, and mechanical characteristics, which render them highly effective in pollutant removal compared to their bulk counterparts [1].

Nanomaterials are successfully applied for diverse environmental applications, including heavy metal adsorption, organic dye degradation, polymer removal, antibacterial activity, and water purification [5]. Their unique properties-high mechanical strength, thermal and electrical stability, quantum confinement effects, and magnetic behaviour-enable advanced environmental functionalities [6]. Additional advantages include lightweight, cost-effectiveness, and ease of mobilization [7]. Analytical detection of hazardous pollutants employs techniques such as high-performance liquid chromatography (HPLC), fluorescence spectroscopy, Raman spectroscopy, Fourier-transform infrared spectroscopy (FT-IR), surface-enhanced Raman scattering (SERS) [8], differential pulse voltammetry (DPV), and cyclic voltammetry (CV). Among these, electrochemical sensing techniques (DPV & CV etc.) have gained prominence due to their high sensitivity, rapid response, real-time monitoring, portability, and cost-effectiveness. Integration of nanomaterials with electrochemical sensors has further enhanced their detection capabilities, providing a robust platform for environmental monitoring [9, 10].

In this review, we provide a comprehensive summary of various classes of advanced nanomaterials and their composites such as metal-organic framework (MOF)-based materials, graphene and graphene quantum dot (GQD)-based nanomaterial, as well as activated carbon (AC) and biochar-derived nanomaterials. The discussion focuses on their application in the electrochemical detection and monitoring of hazardous pollutants in real environmental wastewater samples, highlighting recent advances work for environmental remediation.

2 Types of hazardous compounds

Heavy metals Heavy metals including arsenic (As), lead (Pb), mercury (Hg), cadmium (Cd), and chromium (Cr) are examples of toxic elements that provide serious concerns to human and environmental health because of their toxicity at low concentrations, persistence, and capacity for bioaccumulation [11].

Nitroaromatic compounds (NACs) A class of organic molecules known as nitroaromatic compounds (NACs) has one or more nitro groups ($-NO_2$) joined to an aromatic ring. Trinitrotoluene (TNT), di-nitrotoluene (DNT), and nitrobenzene are typical examples. Manufacturers commonly use these substances to produce medicines, dyes, explosives, and insecticides. NACs are among the most common and worrisome toxic pollutants because of their widespread industrial use and environmental persistence. NACs have a reputation for being extremely poisonous, mutagenic, and possibly carcinogenic. Expo-

sure can happen by ingestion, inhalation, or skin contact, and it can have serious negative health effects, such as harm to the kidneys, liver, and central nervous system [12, 13].

Pesticides In agriculture, pesticides are chemicals that are frequently used to shield crops against diseases, weeds, and pests. Despite, the fact that their application has raised agricultural output, many pesticide chemicals are poisonous to creatures that are not their intended targets, such as beneficial insects, people, and animals. These substances pose significant threats to the environment and human health because they can pollute soil, water, and food chains [14].

Nitrites and nitrates The nitrogen-containing inorganic ions nitrite (NO_2^-) and nitrate (NO_3^-) are frequently found in the environment because of both natural and man-made processes. Industrial operations, animal waste, and sewage discharge produce them, and agriculture also uses them frequently as fertilizer ingredients. Although nitrates and nitrites are necessary for plant nourishment, an excessive buildup of them in water sources can have detrimental effects on human health and the environment [15].

Halo-aromatic compounds (HACs) One or more halogen atoms (such chlorine, bromine, or fluorine) joined to an aromatic ring define halo-aromatic compounds, a family of organic contaminants. These substances are frequently employed as chemical synthesis intermediates, in agriculture (as insecticides and herbicides), cosmetic products and in industrial operations. Polychlorinated biphenyls (PCBs), halogenated phenols, and chlorobenzenes are typical examples. HACs tend to bioaccumulate in the food chain and are persistent in the environment because of their chemical stability, lipophilicity, and resistance to biodegradation. Their capacity to interfere with endocrine function, hinder brain development, and have mutagenic and carcinogenic effects on both people and wildlife [16, 17].

International organizations such as the World Health Organization (WHO), the U.S. Environmental Protection Agency (USEPA), and the European Union Water Framework Directive (EUWFD) establish maximum permissible concentration limits for different pollutants in drinking water and aquatic environments. These limits are often extremely low which are given in Table 1.

Jiokeng et al. shows the functionalization of MOF-808 with histidine to form the MOF-808-His composite. The incorporation of histidine introduces additional nitrogen-rich binding sites, which are expected to enhance the affinity of the material for Pb^{2+} ions via chelation as illustrated in Scheme 1. Then, researchers assessed the electrochemical detection of Pb^{2+} using the modified electrodes [20]. Overall scheme 1 summarize composite functionalization, modification and detection of Pb^{2+} ions.

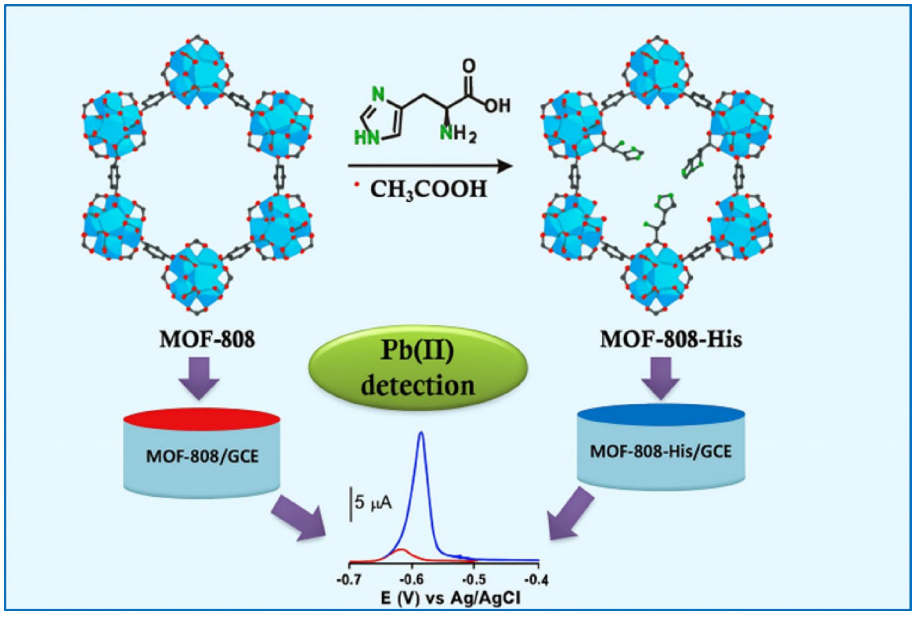
Hira et al. described the synthesis of AgPd@UiO-66-NH_2 and its proposed electrochemical reaction mechanism. Incorporating AgNPs and PdNPs onto the surface of UiO-66-NH_2 forms the bimetallic AgPd@UiO-66-NH_2 composite, which is expected to enhance electron transfer efficiency and catalytic performance. Scheme 2 supports the functional role of the composite in sensing applications by offering a thorough overview of the production of the AgPd-decorated MOF as well as the mechanistic pathway underlying its enhanced electro catalytic activity [26].

Table 1 Maximum permissible concentration of toxic analyte by international organizations

Hazardous compound	Main sources	Environmental risks	Concentration limits (μM)	Health Impacts	References
Pb^{2+}	Industrial activities such as battery manufacturing, mining, plumbing materials etc.	Non-biodegradable heavy metal, leading to long-term contamination of soil and water It accumulates in sediments and aquatic organisms, disrupting microbial activity, plant growth band aquatic fauna	WHO drinking guideline: ≤ 0.048	Hypertension, kidney damage, anemia, reproductive toxicity, bone and neurological disorders	[18–21]
Cd^{2+}	Industrial activities, battery manufacturing, mining, agricultural inputs, waste disposal	Highly toxic, non-biodegradable, easily accumulating in soils, sediments, and aquatic systems.	USEPA drinking guideline: ≤ 0.044	Biological half-life exceeding 10 years, liver damage, anemia, hypertension, severe respiratory irritation, kidney damage human carcinogens, lung and prostate cancers.	[18, 19, 21–23]
Hg^{2+}	Volcanic emissions, forest fires, coal combustion, metal smelting, cement production, domestic, electronic waste		WHO drinking guideline: ≤ 0.005	Nervous system, neurotoxicity, kidney and liver damage cardiovascular disorders, immune system dysfunction	[19, 24, 25]
Cu^{2+}	Industrial discharges such as electroplating, mining, smelting, metal finishing, fertilizer manufacturing, corrosion of copper plumbing systems, runoff from agricultural fields.	Excess copper can disrupt microbial activity in sediments, reducing ecosystem productivity, toxic to aquatic life	WHO drinking guideline: ≤ 0.445	Exposure to high Cu^{2+} levels can cause gastrointestinal irritation, liver and kidney damage, neurological disorders, wilson's disease, oxidative stress and DNA damage in human tissue	[18, 19]
4-NP	Industrial effluents from pesticide and dye manufacturing, petrochemical processing, pharmaceutical industries, and wastewater from textiles	Biodegradation and can accumulate soil and aquatic environments	USEPA drinking guideline: ≤ 0.430	Skin irritation, liver and kidney damage, methemoglobinemia due to interface with oxygen transport in blood, it also affects nervous and endocrine systems, chronic exposure can lead to organ dysfunction and haematological alterations.	[26–28]
NB	Industrial effluents of chemical, pharmaceutical and textile plants, combustion of fossil fuels.	Affecting aquatic life and invertebrate survival and reproduction, slow biodegradation	USEPA drinking guideline: ≤ 0.138	Headache, dizziness, nausea, vomiting, liver, kidney and central nervous system damage, inhalation and dermal contact may result in cyanosis and anemia.	[29–31]
TNT	Military explosives, ammunition factories, waste disposal, contaminated sites	Pollutes soil and water; toxic to fish, plants, and other organisms; long-lasting contamination	USEPA drinking guideline: ≤ 0.008	Skin and eye irritation, anemia, liver damage, possible cancer risk with long-term exposure	[32, 33]

Table 1 (continued)

Hazardous compound	Main sources	Environmental risks	Concentration limits (μM)	Health Impacts	References
NO_2^-	Formed during microbial oxidation of ammonia (nitrification) and reduction of nitrate (denitrification). Major sources include agricultural runoff (fertilizers), industrial and municipal wastewater, animal waste, and sewage effluents	Contributes to aquatic eutrophication and oxygen depletion. Toxic to aquatic life, particularly fish, by inducing methemoglobinemia	USEPA drinking guideline: ≤ 21.730	High levels of nitrite can oxidize haemoglobin to methaemoglobin, reducing oxygen transport in the blood and causing methemoglobinemia ("blue baby syndrome") in infants. Chronic exposure may also lead to hypotension and potential formation of carcinogenic nitrosamines.	[34]
NO_3^-	Agricultural runoff from nitrogen fertilizers and animal manure, Sewage and industrial wastewater, Leakage from septic systems, Atmospheric deposition of nitrogen oxides, Natural soil nitrification processes	Causes eutrophication in aquatic systems, leading to algal blooms and oxygen depletion, contributes to groundwater contamination, Alters soil and water nitrogen balance	WHO drinking guideline: ≤ 806.400	headaches, fatigue, dizziness, and cardiovascular stress in adults at high exposure levels, shortness of breath and blue baby syndrome	[35, 36]



Scheme 1 Histidine grafted MOF for Pb^{2+} detection.
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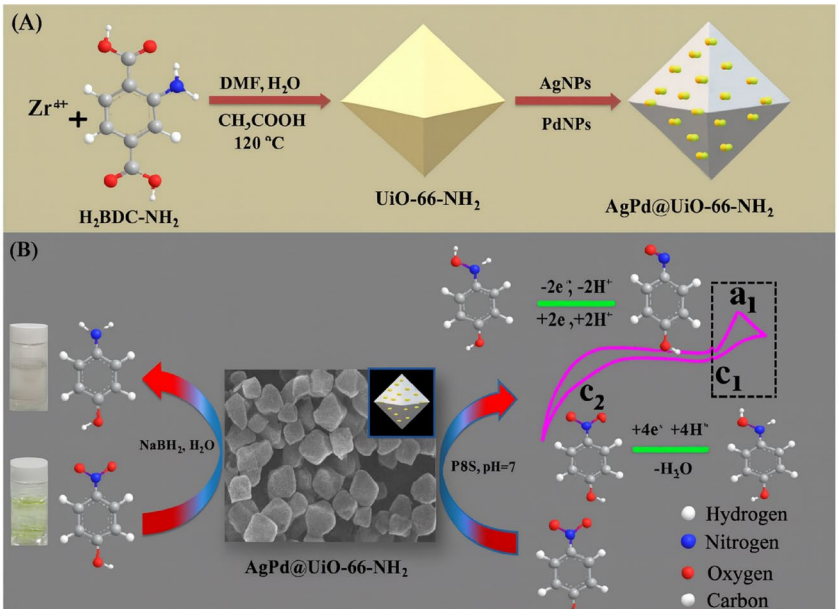

Scheme 2 (**A**) Synthesis of AgPd@UiO-66-NH_2 (**B**) Electrochemical reduction mechanism of *p*-nitrophenol on prepared AgPd@UiO-66-NH_2.
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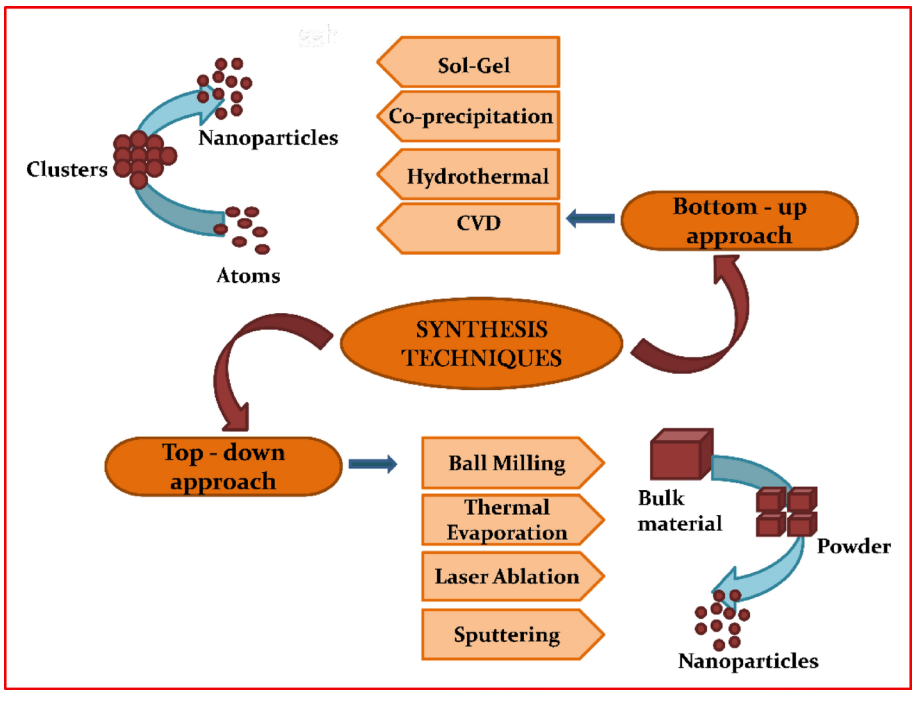

Fig. 1 Synthesis methods of nanomaterials

2.1 Preparation methods of nanomaterials and nanocomposites

Although numerous methods exist to prepare nanomaterials, scientists broadly categorize them into two groups: (i) Top-down approach and (ii) Bottom-up approach (Fig. 1).

Top-down method Through mechanical, physical, or chemical processes, bulk materials are broken down into nanoscale structures in the top-down technique. Using cutting,

milling, or etching procedures, this method begins with a bigger, macroscopic material and reduces it to the nanoscale. Several methods, including laser ablation, etching (both dry and wet), lithography (photolithography and electron-beam lithography), and mechanical milling (ball milling) are used in top-down methods [37]. This approach has several advantages, such as [38–41].

- Scientists prefer top-down processes as they provide excellent alignment, geometric precision, and structural density, which are essential for developing advanced nanodevices.
- Suitable for large-scale production and relatively simple to operate, but it frequently results in uneven particle shapes, surface imperfections, and increased material loss.
- Uses well-established techniques (e.g., ball milling, lithography) that do not require complex chemical conditions.
- Minimal use of solvents or additives, reducing chemical handling.
- Reducing environmental impact, thereby enhancing their sustainable large-scale manufacturing.

Some disadvantages of the top-down approach are [38–42]:

- Limited control over particle size and shape often results in a broad size distribution.
- May introduce defects, impurities, or structural damage during mechanical processing.
- Difficult to produce tiny (< 10 nm) particles without aggregation or damage.
- Processes can be energy intensive and generate waste, raising environmental concerns.

Bottom-up method Using chemical or biological processes, the bottom-up technique assembles nanostructured materials from atomic or molecule building blocks. By simulating natural material creation processes, this technique allows for exact control over surface functioning, crystallinity, particle size, and shape [43]. The bottom-up approach builds nanomaterials from atoms or molecules through chemical or biological synthesis routes such as sol–gel, hydrothermal synthesis, precipitation, chemical vapour deposition (CVD), and self-assembly [40]. This approach has advantages, such as.

- Allows precise control over particle size, shape, and composition by adjusting reaction conditions.
- Enables tailoring of structural and functional properties, useful for advanced materials like quantum dots and core–shell systems.
- Can be more environmentally friendly when using green synthesis routes (e.g., plant extracts, biological reducers).
- Well-suited for creating complex or functionalized nanocomposites for catalysis, drug delivery, and sensing.
- Cost-effectiveness, as they rarely require expensive or sophisticated equipment.

Some disadvantages of bottom-up approaches are [40, 41, 44]:

- Requires precise control of reaction parameters; slight errors can affect product quality.

- Often more expensive because of high-purity precursors and specialized synthesis equipment.
- Reactions may require a longer time and are highly sensitive to impurities.
- Demands trained personnel and advanced instrumentation (e.g., for CVD, sol–gel systems).

2.2 Advanced electroactive nanomaterials

Metal-organic frameworks (MOFs) (Fig. 2) MOFs are a type of hybrid porous material made up of organic linkers and metal nodes. MOFs are promising options for addressing the demands of upcoming energy storage systems [45]. Metal-based modifiers play a pivotal role in augmenting the performance of electrochemical sensors by significantly enhancing their sensitivity, selectivity, and overall electrochemical responsiveness [46]. The MOF's based nanomaterials are crystalline with less than 2 nm or 2–50 nm, mesoporous structure. These materials include several properties, such as considerable chemical diversity and large surface area [47]. There are several methods available for preparing MOFs such as diffusion synthesis, hydrothermal or solvothermal, mechanochemical, spray-drying, ultrasonic/microwave, and electrochemical [48].

Metal-organic frameworks (MOFs) exhibit exceptional potential for the detection of heavy metal ions due to their precisely tunable porosity, exceptionally high specific surface area, and the inherent chemical versatility of their metal-ligand coordination environments. The presence of coordinatively unsaturated metal nodes (e.g., Cu^{2+}, Fe^{3+}, Zr^{4+}) provides active binding sites capable of establishing strong and selective interactions with electron-donating heavy metal ions such as Pb^{2+}, Cd^{2+}, and Hg^{2+} through hard soft acid-base coordination. Moreover, the physicochemical nature of the organic

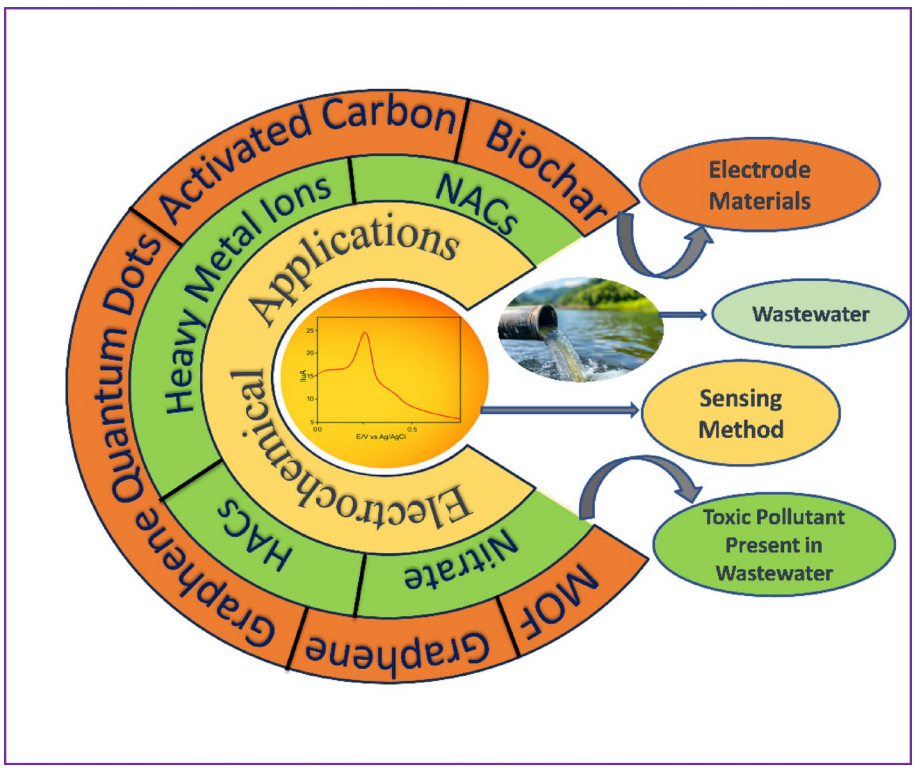

Fig. 2 Schematic illustration of advanced electroactive nanomaterials and application for wastewater monitoring

linker based on carboxylate, imidazolate, or phosphonate functionalities plays a pivotal role in dictating the framework's, binding affinity and ion selectivity. Specifically, linkers enriched with soft donor atoms such as sulfur or nitrogen facilitate soft acid–soft base interactions, thereby enhancing the affinity toward Hg^{2+}, while oxygen-donor-rich linkers preferentially coordinate with harder metal centres such as Pb^{2+} [49–51].

Graphene based nanomaterials/ graphene quantum-dots Since its discovery in 2004 as a two-dimensional (2D) monolayer carbon nanomaterial. Because of its remarkable electrical, mechanical, and thermal capabilities, graphene a single layer of sp^2 hybridized carbon atoms organized in a two-dimensional honeycomb lattice has garnered a lot of interest lately. Graphene is widely used in electrochemical sensing platforms to improve sensitivity, selectivity, and response time because of its high surface area, superior conductivity, and potent π–π interactions [52]. These characteristics make it perfect for promoting quick electron transfer reactions, which are essential for identifying a variety of analytes, such as environmental contaminants, heavy metals, and biomolecules [53].

Graphene and its derivatives, including graphene oxide (GO) and reduced graphene oxide (rGO), have emerged as highly efficient transduction materials for the electrochemical detection of nitroaromatic compounds. Their superior sensing performance can be attributed to the extended π-conjugated carbon lattice, exceptional electron mobility, and high density of surface-active sites that collectively facilitate rapid charge transport and molecular adsorption. Nitroaromatic analytes such as nitrobenzene and 2,4,6-trinitrotoluene (TNT) possess electron-deficient aromatic rings owing to the presence of strongly electron-withdrawing nitro substituents. These moieties engage in π–π stacking and donor–acceptor charge-transfer interactions with the delocalized π-electron cloud of graphene-based materials, thereby enhancing interfacial electron transfer kinetics and amplifying the electrochemical signal. Consequently, graphene-derived nanostructures serve as ideal conductive matrices for the sensitive and selective detection of nitroaromatic pollutants [54–56].

GQDs are tiny pieces of graphene sheets, usually less than 10 nm. They combine the special qualities of graphene with edge effects and quantum confinement. Because of their exceptional biocompatibility, superior water dispersibility, and programmable photoluminescence, GQDs are very helpful in electrochemical and biosensing applications [57, 58].

Activated carbon/ Biochar Both AC and biochar are carbon-rich porous materials that are frequently utilized in sensing and environmental remediation technologies. Despite certain structural and chemical similarities, their physicochemical characteristics, preparation techniques, and origins set them apart in terms of their performance and uses in electrochemical detection platforms [59]. Biochar (BC) is a carbonaceous substance that is created when biomass such as sewage sludge, forestry waste, and agricultural residues-is pyrolyzed in an oxygen-limited environment. By modifying the feedstock type and pyrolysis parameters (temperature, heating rate, and residence time), its characteristics, such as surface area, pore size distribution, and surface functional groups, may be adjusted [60]. BC has abundant availability, low cost, sustainability and adaptability make it a superior choice electrochemical sensing application [61].

The extremely porous carbon substance known as AC is made by chemically or thermally activating carbonaceous precursors. Rapid analyte diffusion and adsorption are made possible by its hierarchical pore structure and remarkably large surface area (up to 2500 m^2/g). The design of electrodes for electrochemical sensors and supercapacitors has long made use of AC because of its high conductivity, chemical stability, and variable surface chemistry [62].

Biochar and activated carbon (AC) are two different but closely related groups of carbonaceous materials that are important for the development of sustainable sensors [63, 64]. Activated carbon is known for its large surface area, well-developed porosity, and exceptional adsorption capacity. By reducing the production of hazardous by-products, the use of physical activation (steam or CO_2) rather than chemical activating agents (e.g. KOH, $ZnCl_2$) further improves environmental compatibility [65]. However, biochar is produced directly from the pyrolysis of biomass wastes with little processing needed, it provides an intrinsically renewable and carbon-negative substitute. Surface activation and heteroatom doping can greatly increase its electrochemical activity without sacrificing sustainability, even if its intrinsic conductivity is smaller than that of AC [66, 67] (Table 2).

2.3 Electrochemical sensors and their fabrication

Commonly utilized working electrodes in electrochemical sensing systems include glassy carbon electrodes (GCE), gold electrodes (GE), Fluorine-doped tin oxide (FTO) electrode and screen-printed electrodes (SPEs), owing to their excellent conductivity, chemical stability, and surface modifiability. Quantitative determination of analyte concentrations was typically achieved using electrochemical techniques such as cyclic voltammetry (CV), differential pulse voltammetry (DPV), square wave voltammetry (SWV), and amperometry (AMP), each offering distinct advantages in terms of sensitivity, resolution, and analytical precision. In the electrochemical detection of toxic analyte, the sensing mechanism predominantly relies on redox processes occurring at the

Table 2 Synthesis methods of different electroactive nanomaterials and types of pollutants

Materials	Synthesis methods	Performance rationale	Typical target pollutants	References
Metal-Organic Frameworks (MOFs)	Solvothermal / hydrothermal synthesis, Microwave-assisted, electrochemical, spray-drying, and diffusion synthesis.	Controlled crystal structural, tuneable porosity, high surface area for adsorption, stability, particle morphology and conductivity.	Heavy metals ion (Pb^{2+}, Cd^{2+}, Hg^{2+}), dyes, antibiotics.	[22, 45, 48, 68, 69]
Graphene-based Nanomaterials	Chemical exfoliation, Hummer's method, Reduction of graphene oxide (rGO), Chemical Vapour Deposition.	High conductivity, large surface area, hydrophilic nature, high functional group density, good chemical stability.	Phenols, pesticide, dyes, explosives.	[29, 32, 52, 53]
Graphene Quantum Dots (GQDs)	Hydrothermal reduction, Bottom-up synthesis, chemical exfoliation, electron beam lithography.	Quantum confinement provides high photoluminescence and strong signal for fluorescence/electrochemical detection	Heavy metals ion, pharmaceuticals, nitrites/nitrates	[57, 70, 71, 72]
Activated Carbon/ Biochar	Biomass pyrolysis conditions, Chemical/steam activation, Hydrothermal carbonization	Low-cost porous, high specific surface area, large porosity, stability, electrical conductivity.	Dyes, organic pollutants, explosives, phenols, industrial effluents	[60, 62, 73]

working electrode (WE) surface, where oxidation and reduction reactions of the target analytes generate measurable electrical signals proportional to their concentrations [74].

Despite the significant progress in the development of electroactive materials for water remediation, the current body of literature remains fragmented in scope and analytical depth. Most existing reviews have primarily emphasized either the synthesis of specific material such as carbon-based nanocomposites and transition-metal oxides, or their individual electrochemical mechanisms. However, there remains a lack of integrative analysis i.e. toxicity nature of analyte by internation organisation (WHO & USEPA), materials classification, material structure, environment sustainability of some materials and high % recovery rate. Furthermore, few studies have examined the interplay between electrochemical activity, stability under realistic water matrices, and scalability for continuous treatment operations. This review aims to bridge these gaps by providing a comprehensive and comparative assessment of diverse electroactive materials, focusing on their structure-property-function relationships, and application challenges. By critically analysing existing knowledge and highlighting current issues, this work establishes a coherent framework to guide future research toward sustainable and high-performance electroactive systems for environmental remediation via electrochemical method.

2.4 Use of advanced nanomaterials for electrochemical applications in wastewater monitoring

2.4.1 Use of MOF based nanomaterial

In 2019, Hira et al. reported Silver-palladium (AgPd) bimetallic nanoparticles were incorporated into functionalized MOF as a stable host matrix to producing a highly effective electrocatalytic platform. FTIR, XRD, SEM, TEM, XPS, thermogravimetric analysis (TGA), Brunauer-Emmett-Teller (BET) surface area analysis, and UV-Visible spectroscopy were used to characterize the MOF. With a remarkable limit of detection (LOD) of around 32 nM (nano molar) and a broad linear detection range of 100–370 μM, the electrochemical sensor that was created showed excellent sensitivity and selectivity for 4-NP. The practical usefulness of sensor's done by detection of 4-NP in tap water samples, with a high recovery rate of about 100.97% [26]. Chen et al. published square wave anodic stripping voltammetry (SWASV) in 2021 using an electrode based on a metal–organic framework (MOF). ZIF-67 was used as the starting substrate for the sensing interface, and then Fe, Mn, or Ni ions were added utilizing a simple doping procedure with NaOH assistance to create bimetallic MOF nanocomposites. By drop-casting the resultant nanostructures onto a glassy carbon electrode (GCE), the detection performance of Hg^{2+} was significantly improved. Additionally, tests using electrochemical impedance spectroscopy (EIS) and CV validated the Fe-doped MOF nanocomposite's enhanced charge transfer capabilities. The sensor reported spike recovery rates above 99.00% for Hg^{2+} concentrations in the range of 0.3–1.0 μM and showed outstanding operational stability during 10 consecutive measurement cycles [75]. In 2021, *Singh et al.*, described a copper-based metal-organic framework (Cu-MOF) and multi-walled carbon nanotube (CNT) nanocomposite that was made using a one-pot hydrothermal approach for the quick and economical detection of Cd^{2+} ions. XRD, FT-IR, BET, SEM, TEM, and other techniques were used to characterize the structure and morphology. In 0.1 M phosphate buffer, the composite was immobilized onto a glassy carbon electrode (GCE) for electrochemical detection. Under ideal circumstances, the sensor

demonstrated a broad linear calibration range of 0.2–10 μM and a low detection limit of 0.275 nM (S/*N* = 3). The electrode modified with CNT-Cu-MOF also showed outstanding stability, repeatability, and selectivity. The sensor's practical applicability was confirmed by detecting Cd^{2+} in real tap water samples, achieving a recovery of 100.40% [22]. Chen et al. reported in 2021, even at trace levels, the presence of Cd^{2+}, Pb^{2+}, Cu^{2+}, and Hg^{2+} in drinking water poses serious health hazards, highlighting the necessity for sensitive detection techniques. Here, we describe a one-pot synthesis of amino-functionalized metal-organic frameworks (NH_2-MIL-88(Fe)) embedded with acid-functionalized carbon nanotubes (CNTs) and graphitic carbon nitride (*g*-C_3N_4) nanosheets. The prepared nanocomposite was coated in GCE and square wave stripping voltammetry (SWASV) technique was applied. The developed sensor has sensitivities of 0.0789, 0.4122, 0.2616, and 0.3251 mA μM^{-1} cm^{-2}, respectively, and detection limits of 39.6 nM (Cd^{2+}), 7.6 nM (Pb^{2+}), 11.9 nM (Cu^{2+}), and 9.6 nM (Hg^{2+}). The sensor was successfully applied in real water samples [76].

In 2022 Hu et al. reported chromium-based metal–organic framework (Cr-MOF) nanoparticles using a simple hydrothermal method. Nanoparticle was characterised by X-ray diffraction (XRD), TEM, and SEM to confirm their crystalline structure and morphology. For the detection of 4-nitrophenol (4-NP), a Cr-MOF-modified glassy carbon electrode (Cr-MOF/GCE) was successfully created and used as an extremely effective electrochemical sensing platform. Sensitive detection was made possible by the Cr-MOF nanoparticles' exceptional electrocatalytic activity toward the electrochemical reduction of 4-NP. The prepared sensor showed a very low detection limit of 0.7 μM and a wide linear detection range of 2–500 μM. the Cr-MOF/GCE platform as a viable option for nitro-phenolic pollutant trace-level detection in local river water samples [68]. In 2023, Luo et al. reported the development of a highly sensitive electrochemical sensor based on metal-organic framework-derived carbon/molybdenum disulfide (MOFs-C/MoS_2) nanocomposites for the detection of 4-NP in wastewater and lake water samples. The nanocomposite material was synthesized via a solvothermal method. Square wave voltammetry (SWV), a highly sensitive electrochemical technique, was employed for the quantitative detection of 4-NP, achieving a broad linear detection range of 0.5–500 μM [27]. Ambaye et al. 2023, reported the detection of chemical pollutants such as 2,4-dichlorophenol (2,4-DCP) from wastewater sample by using electrochemical sensing method. The Fe-MOF-based g-C_3N_4 nanocomposite was coated on a barescreen-printed carbon electrode (SPCE) to detect the 2,4-DCP and the LOD was found to be 1.2 nM with a linear range 0.5–200 μM. The nanocomposite material was synthesized *via* the solvothermal method and characterized by FESEM, EDS, HR-TEM, XRD, FTIR, XPS, TGA/DSC, CV and EIS [77]. In 2023, Tran et al. developed an electrochemical sensing platform that uses MIL-88B(Fe)-NH_2, an amine-functionalized metal-organic framework (MOF), as a surface modification for a GCE to detect numerous heavy metal ions in aquatic environments simultaneously. By acting as efficient binding and preconcentration sites for target metal ions, the many amine groups in MIL-88B(Fe)-NH_2 improve electrochemical oxidation signals and detection sensitivity. XRD, SEM, TEM, FTIR, CV, and EIS were used to thoroughly characterize MIL-88B(Fe)-NH_2, which was synthesized using a solvothermal method under ideal conditions. Outstanding electroanalytical performance is demonstrated by the manufactured sensor, with broad linear detection ranges of 0.025–1.000 μM for Cd^{2+}, 0.30–10.0 μM for Pb^{2+}, and 0.60–10.0 μM

for Cu^{2+} and limits of detection (LOD) of 0.2 nM, 192 nM, and 381 nM, respectively. Additionally, the sensor demonstrates outstanding stability, selectivity, and repeatability, highlighting its potential as a flexible and effective method for quantitative measurement and real-time monitoring of heavy metal ion pollutants in environmental water samples [18]. A new sulfur-bridged metal-organic framework (MOF) based on thiacalix [4]arene derivatives, $[Co(TIC_4R\text{-}I)_{0.25}Cl_2]\cdot 3CH_3OH$ (Co-TIC_4R-I), was successfully synthesized by Guo et al. in 2023 using a solvothermal technique. In this framework, a strong three-dimensional (3D) microporous architecture is formed via the interconnection of neighbouring TIC_4R-I ligands via Co^{2+} nodes. Heavy metal ions (HMIs), such as Cd^{2+}, Pb^{2+}, Cu^{2+}, and Hg^{2+}, may be detected in aqueous solutions using a highly sensitive electrochemical sensor that was prepared by immobilizing the synthesized Co-TIC_4R-I onto a glassy carbon electrode (Co-TIC_4R-I/GCE). With LODs of 0.017 μM, 0.008 μM, 0.016 μM, and 0.007 μM, the Co-TIC_4R-I/GCE sensor showed broad linear detection ranges of 0.10–17.00 μM, 0.05–16.00 μM, 0.05–10.00 μM, and 0.80–15.00 μM for Cd^{2+}, Pb^{2+}, Cu^{2+}, and Hg^{2+}, respectively. The Co-TIC_4R-I/GCE-based sensor provides a reliable and effective method for determining many heavy metal ions at the trace level in aquatic system [19] (Table 3).

In 2023, Garg et al. reported employing SWASV to selectively and easily detect trace quantities of cadmium (Cd^{2+}) electrochemically using an iron-based metal–organic framework (MOF)-modified GCE. To improve sensitivity and selectivity, NH_2-MIL-53(Fe), a functionalized Fe-based MOF, was applied to the electrode surface to create the sensing interface. For the detection of Cd^{2+}, the NH_2-MIL-53(Fe)/GCE sensor showed a broad linear response range of 44 nM-2.22 μM, with an estimated LOD of 0.03 μM and a limit of quantification (LOQ) of 0.09 μM. The effective identification of Cd^{2+} in spiked water samples further confirmed the proposed sensing platform's practical usability [23]. According to He et al. (2023), it is crucial to detect Cd^{2+} in environmental samples quickly and sensitively. He works, in highly sensitive dual-hairpin (HP) electrochemical aptasensor for Cd^{2+} ion detection using Au/Cu_2O as a functional substrate and AuPd/Fe-MOF nanocomposites as signal labels. Energy-dispersive spectroscopy (EDS), XRD, and SEM were used to characterize the nanocomposite and establish its shape and elemental distribution. CV and EIS were used to investigate the sensor's electron transfer characteristics and construction procedure, and DPV was used to quantitatively assess the Cd^{2+} detection capability. DPV showed an ultralow detection limit of 2.27×10^{-5} μM and a broad linear detection range of 10^{-4}–10 μM for Cd^{2+} in real water samples [78]. A new electrochemical sensing platform for the detection of heavy metal ions in aqueous solutions was published by Doloi et al.. in 2024. The platform uses an indium tin oxide (ITO) electrode modified with a zirconium-based metal-organic framework (MOF, UiO-66) decorated with Ag_2O nanoparticles (NPs). Using differential pulse stripping voltammetry (DPSV), the Ag_2O/UiO-66-modified electrode allowed for the independent and simultaneous detection of Hg^{2+} and Cd^{2+}. High sensitivities of 551 μA μM^{-1} cm^{-2} for Cd^{2+} and 341.2 μA $^{-1}$ cm^{-2} for Hg^{2+}, with corresponding detection limits of 0.008 μM and 0.003 μM. The method also shows excellent selectivity, sensitivity, and repeatability, so it can be used for real-world environmental monitoring of heavy metal pollution in groundwater, wastewater, soil, and river water [69]. The new electrochemical sensor for Pb^{2+} ion detection was described by Jiokeng et al. in 2025. It is based on a GCE that has been modified with a thin layer of histidine-functionalized metal-organic framework

Table 3 Electroactive advanced nanomaterials monitoring of wastewater

SN	Material	Pollutant	Technique	LR (μM)	LOD (μM)	Interference	References
1.	AgPd@UiO-66-NH_2/GCE	4-NP	DPV	100–370	0.032	2-NP, 3-NP, phenol, nitrobenzene, hydroquinone, catechol, resorcinol, 2,4-dinitrophenol (DNP), 2-AP, 2,4,6-trinitrophenol (TNP)	[26]
2.	Fe1Co1/GCE	Hg^{2+}	SWASV	0.1–1.1	0.00782	Pb^{2+}, Cd^{2+}, Cu^{2+}	[75]
3.	CNT-Cu-MOF/GCE	Cd^{2+}	DPV	0.2–10	0.00027	Pb^{2+}, Cd^{2+}, Cu^{2+}, Hg^{2+}, Ni^{2+}, Zn^{2+}, Co^{2+}, Mn^{2+}	[22]
4.	NH_2-MIL-88(Fe)@ g-C_3N_4/GCE	Cd^{2+}, Pb^{2+}, Cu^{2+}, Hg^{2+}	SWSV		0.039, 0.0076, 0.0119, 0.0096		[76]
5.	Cr-MOF/GCE	4-NP	DPV	2.0–500	0.7	$MgCl_2$, KCl, $CaCl_2$, NaCl, Na_2SO_4, HQ, phenol, catechol	[68]
6.	MOFs-C/MoS_2/GCE	4-NP	SWV	0.5–50	0.04	KCl, NaAc, $MgSO_4$, Na_2CO_3, KNO_3, phenol, resorcinol	[27]
7.	Fe-MOF/g-C_3N_4/ SPCE	2,4-DCP	DPV	0.5–15	0.00120	sodium nitrate, ascorbic acid, 4-NP, 2-NP, bisphenol A, phenol, 4-chlorophenol, 2-chlorophenol, sodium sulphate decahydrate, Co^{2+}, Pb^{2+}, Sr^{2+}, Cr^{2+}, Cu^{2+}, Mo^{2+}, Fe^{2+}, Li^{2+}, Mg^{2+}, Mn^{2+}, Sb^{2+}, Se^{2+}, Ni^{2+}, Ti^{2+}, V^{2+}, Zn^{2+}	[77]
8.	MIL-88B(Fe)-NH_2/GCE	Cd^{2+} Pb^{2+} Cu^{2+}	DPV	0.025–1.00, 0.3–10.00, 0.6–10.00	0.00020 0.192 0.381	Co^{2+}, Ni^{2+}, Zn^{2+}, $Cr_2O_7^{2-}$, K^+, Cl^-	[18]
9.	Co-TIC4RI/GCE	Cd^{2+} Pb^{2+} Cu^{2+} Hg^{2+}	SWASV	0.10–17.00, 0.05–16.00, 0.05–10.00, 0.80–15.00	0.017 0.008 0.016 0.007	Bi^{3+}, As^{3+}, $Ni2^+$, Mg^{2+}, Zn^{2+}, Ag^+, K^+, Ca^{2+}, Na^+	[19]
10.	NH_2-MIL-53(Fe)/GCE	Cd^{2+}	SWASV	0.0448–2.22	0.03	Pb^{2+}, Zn^{2+}, As^{5+}, Se^{6+}	[23]
11.	AuPd/Fe-MOF	Cd^{2+}	DPV	0.0001–8.90	0.000022	Mn^{2+}, Zn^{2+}, Pb^{2+}	[78]
12.	S1(MOF)/ITO S2(MOF)/ITO	Cd^{2+} Hg^{2+} Cd^{2+} Hg^{2+}	DPSV	0.05–1.00, 0.01–1.20, 0.05-1.00, 0.01–1.20	0.05 0.02 0.008 0.003	Zn^{2+}, Pb^{2+}, Cl^{2+}, Fe^{2+}	[69]
13.	MOF-808-His/GCE	Pb^{2+}	SWASV	0.001–0.5	0.00011	Cd^{2+}, Cu^{2+}, Zn^{2+}, Cr^{3+}, Sn^{2+}, Mn^{2+}, Ni^{2+}, Ag^+, Mg^{2+}, In^{3+}, Tm^{5+}	[20]
14.	EDTA-NH_2-UiO-66/G/ GCE	Pb^{2+} Hg^{2+}	DPASV	0.1–50	0.085 0.083	Cu^{2+}, Cd^{2+}, Na^+, K^+, Cr^{3+}, Fe^{3+}, Fe^{2+}	[79]

Table 3 (continued)

SN	Material	Pollutant	Technique	LR (μM)	LOD (μM)	Interference	Referenes
15.	CMF-RGO/GCE	NB	DPV	0.2–927.7	0.088	4-NT, 4-NP, 4-C-NB, 4-b-NB, 4-Nb acid, Cd^{2+}, Hg^{2+}, Zn^{2+}, Ca^{2+}	[29]
16.	Graphene nano-sheets/CPE	Chloroxylenol	SW-AdASV		0.00137		[80]
17.	TRGOPE	Chloroxylenol	SWV	0.1–4.5	0.0208		[81]
18.	GO/GCE	NB	CV, LSV, DPV, SWV	-	0.066, 0.146, 0.079, 0.135	Mg^{2+}, Cd^{2+}, Pb^{2+}, Hg^{2+}, Na^{+}, K^{+}, Phenol, catechol, hydroquinone, resorcinol	[30]
19.	rGO-MWCNT/BDD	TNT	SWV	0.5–1100	0.019	-	[32]
20.	ZnO@G/SPCE	Cd^{2+} Pb^{2+}	ASV	10–200	0.00534 0.00386	Na^{+}, K^{+}, Ca^{2+}, Mg^{2+}, Ba^{2+}, Cu^{2+}, Co^{2+}, Ni^{2+}, Zn^{2+}, Mn^{2+}, Fe^{3+}, Cl^{-}, SO_4^{2-}, NO^{3-}	[82]
21.	NCQDs-GO/GCE	Cd^{2+} PB^{2+}	ASV	0.100–100 0.100–50	0.0663 0.00565	K^{+}, Na^{+}, Ni^{2+}, Fe^{3+}, Mg^{2+}, Mn^{2+}, Zn^{2+}, Hg^{2+}, Cu^{2+}	[83]
22.	Ag/AEfG/GCE	NO_2^{-}	Amperometry	0.05–3000	0.023	KCl, $NaNO_3$, $MgSO_4$, $CaCl_2$, nitrobenzene, 4-nitrotoluene, 4-nitrochlorobenzene.	[72]
23.	Rh-rGO/GCE	NO_3^{-}	CV				[36]
24.	Au/rGOCNT10-1/Bi modified E	Cd^{2+} Pb^{2+}	SWASV		0.000534 0.0000965	K^{+}, Na^{+}, Ca^{2+}, Cl^{-}, NO^{3-}, Fe^{3+}	[84]
25.	N-GQD/GCE	TNP	DPV	0.001–100	0.001	Hg^{2+}, Ag^{+}, Pb^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+}, Zn^{2+}, Sn^{4+}, Cu^{2+}, Cr^{6+}	[85]
26.	GO-Fe_3O_4-PAMAM/GCE	Pb^{2+} Cd^{2+}	SWASV	0.0019–0.579, 0.0017–1.25	0.00062 0.00062	Tl+, Cu^{2+}, Zn^{2+}, Hg^{2+}, Co^{2+}, Mg^{2+}, Ca^{2+}, Mn^{2+}, Ni^{2+}, Fe^{3+}, In^{3+} Cr^{3+}	[86]
27.	N-doped GQDs/ITO	Hg^{2+}	CV	0.05–0.25	0.05	Cd^{2+}, Zn^{2+}, Cu^{2+}, Ni^{2+}, Mg^{2+}, Co^{2+}, Ba^{2+}, Mn^{2+}, Pb^{2+}, Cr^{2+}, Ca^{2+}, K^{+}, Al^{3+}, Fe^{3+}, Hg^{2+}	[24]
28.	GO/MgO/GCE	NB	DPV	0.1–38.9 and 58.5–2333.5	0.01	4-NP, 4-NA, Pb^{2+}, Cd^{2+}, Na^{+}, Ca^{2+}, Cu^{2+}, Br^{-}, Cl^{-}	[31]
29.	TiO_2GO-1/GE, TiO_2GO-2/GE	NB	CV	0.016–0.065 0-0.032	0.021		[87]
30.	SIGNS/GCE	TNT	DPV	-	0.00528	NB, 2-NT, 4-NT, 4-DNT, 2, 4, 6-NT	[88]
31.	N-GQD/GCE	Cd^{2+}	DPV	0.0000089–997	8.9×10^{-8}	Cd^{2+}, Hg^{2+}, Ag+, Pb^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+}, Zn^{2+}, Sn^{2+}, Cu^{2+}, Cr^{4+}	[89]

Table 3 (continued)

SN	Material	Pollutant	Technique	LR (μM)	LOD (μM)	Interference	Referenes
32.	GQDTU-IIP-GCE	Hg^{2+}	DPV	0.05–23.0	0.0235	Pb^{2+}, Cd^{2+}, Cu^{2+}, Ni^{2+}, Mn^{2+}, Co^{2+}, Zn^{2+}, As^{3+}, Al^{3+}, Fe^{2+}	[90]
33.	GQD-DPA/GCE	Pb^{2+}	DPASV	0.00003–0.0028	1.0×10^{-5}	Zn^{2+}, Cd^{2+}, Ni^{2+}, Co^{2+}, Cu^{2+}	[91]
34.	PEDOT-PSS/rGO	Pb^{2+}	DPASV	0.00483-0.338	0.000435	Hg^{2+}, Cd^{2+}, Cu^{2+}, Zn^{2+}, Na+, Fe^{3+}	[92]
35.	prGO-MoS_2/GCE	Pb^{2+} Cd^{2+}	DPV	0.0000001–1000, 0.0001–10	4.8×10^{-7} 4.4×10^{-2}	Hg^{2+}, Ag+, Fe^{2+}, Co^{2+}, Ni^{2+}, Zn^{2+}, Sn^{2+}, Cu^{2+}, Cr^{4+},	[93]
36.	S, N-GQD/GCE	Cd^{2+} Pb^{2+} Hg^{2+}	DPV	0.0001–100	0.000001 0.00001 0.000001	Ag+, Fe^{3+}, Fe^{2+}, Co^{2+}, Ni^{2+}, Zn^{2+}, Mg^{2+}, Li^{2+}, Mn^{2+}, Sn^{2+}, Cu^{2+}, Cr^{4+}	[94]
37.	MoS_2-rGO/CPE	Hg^{2+}	CV	0.1–10	1.6	Pb^{2+}, Cd^{2+}, Ca^{2+}, Na^{+}, Mg^{2+},	[95]
38.	PTH-afGQDs/GCE	Hg^{2+}		0.000001–1.0	6.0×10^{-7}	K^{+}, Na^{+}, Pb^{2+}, Cd^{2+}, Al^{3+}, Ni^{2+}, Zn^{2+}, Cr^{3+}, Mn^{2+}, Fe^{2+}, Fe^{3+}	[70]
39.	SrTiO3/Ag/rGO/ SPCE	4-NP	LSV	0.1–1000	0.00003	Mg^{2+}, Cu^{2+}, Ca^{2+}	[71]
40.	$(AgNPs\text{-}rGO)_{TL}$:g-C_3N_4/GCE	4-NP	DPV	0.015–8001	0.015	Gl, Ur, CA, BSA, HQ, CC, 4-AP	[72]
41.	Hyb-BNQD-NGQD/GCE	Pb^{2+}	DPV	10^{-6}–10^{2}	1.0×10^{-6}	Na^{+}, Mg^{2+}, Cd^{2+}, Co^{2+}, Li^{+}, Cr^{3+}, Ca^{2+}, K^{+}, Hg^{2+}, Ni^{2+}, Zn^{2+}, Mn^{2+}, Cu^{2+}	[96]
42.	Pd@PAC/GCE	Cd^{2+}, Pb^{2+}, Cu^{2+}	SWASV	0.025–0.500	0.0133, 0.0066, 0.0119	Cd^{2+}, Pb^{2+}, Cu^{2+}	[97]
43.	Au-rGO/AC/GCE	4- NP	LSV, Amperometry	72–4400, 18–592	1.12, 0.36	KCl, $MgSO_4$, Na_2SO_4, K_2CO_3, $CaCl_2$, $CoNO_3$, Ur, Oxalic acid, Gl, Sucrose, Fructose, Ascorbic Acid, Dopamine, Uric Acid, Thiourea, 3-NP, Nitrobenzene, 4-Nitrotoluene	[28]
44.	AC-rGO/CPE	Pb^{2+}, Cd^{2+}	CV, LSV	0.0048–0.362, 0.0089–0.667	0.0527, 0.125		[21]
45.	ACOP/MWCNT/CPE	Cd^{2+}	SWV	9.09–72.9	0.910		[98]
46.	Bi/A-GCE	4-NP	DPV	0.005-170	0.00018	2-NP, metronidazole, acetates, SO_4^{2-}, nitrates, CO_3^{2-}, hydrocortisone, AA, Gl	[73]
47.	BC/Au/FTO	Nitrite	DPV	0.5–6000	0.14	SO_4^{2-}, CO_3^{2-}, K^{+}, Fe^{3+}, Zn^{+}, Ca^{2+}, Mg^{2+}	[34]

Table 3 (continued)

SN	Material	Pollutant	Technique	LR (μM)	LOD (μM)	Interference	Referenes
48.	BC/Cu_2O/GCE	Hg^{2+}	DPASV	4.98×10^{-6} − 4.98	1.4×10^{-6}		[25]
49.	BieneNS@C/GCE	Cd^{2+}	DPV	0–1.34	0.0018	Ca^{2+}, Co^{2+}, Fe^{3+}, NH_4^+, Zn^{2+}, Pb^{2+}, Cu^{2+}	[99]

GCE: Glassy carbon electrode, CMF-RGO: Cellulose microfiber/reduced graphene oxide, G: Graphene, GO/MgO: Graphene oxide/magnesium oxide, rGO-MWCNT: reduced graphene oxide/multi-walled carbon nanotube, BDD: boron-doped diamond, SIGNS: Si doped graphene nanosheets, GO-Fe_3O_4-PAMAM: Poly(amidoamine) dendrimer functionalized magnetic graphene oxide, prGO-MoS_2/GCE: Partially reduced graphene oxide-nano molybdenum disulphide-modified glassy carbon electrode, N-doped GQDs: Nitrogen doped graphene quantum dots, Hyb-BNQD-NGQD: Hybrid Boron Nitride/N-Doped graphene Quantum Dots, ZnO@G/SPCE: Zinc Oxide Graphene /Screen-printed carbon electrode, NCQDs-GO: N-doped carbon quantum dots-graphene oxide, N-GQD/GCE: Nitrogen-doped graphene quantum dots, S,N-GQD: Sulphur co-doped Nitrogen-graphene quantum dot, GOCS: Carbon disulphide-functionalized graphene oxide, MoS_2-rGO: Molybdenum disulphide and reduced graphene oxide, GCE/PTH-e GQDs: Poly(thionine).Amino-functionalized graphene quantum dots, GQDTU-IIP-GCE: Thiourea derivative functionalised graphene quantum dot-Ion Imprinted polymer, GQD-DPA/GCE: D-penicillamine Graphene quantum dot, PEDOT-PSS/rGO: Poly(3,4-ethylenedioxythiophene): poly(sterene sulfonate)/reduced graphene oxide, Au/rGOCNT10-1/Bi: micro-patterned reduced graphene oxide/carbon nanotube/bismuth composite, Ag/AEfG: Ethanolamine and Ag-nanoparticle dual-functionalized graphene oxide, Rh-rGO: Rhodium –reduced graphene oxide, Gl: Glucose, Ur: Urea, HQ: Hydroquinone, CC: Catechol, AA: Ascorbic acid DPV: Difference pulse voltammetry, CV: Cyclic voltammetry, LSV: Linear sweep voltammetry, SWV: square wave voltammetry, SWASV: Square wave anodic stripping voltammetry, DPASV: Differential pulse anodic stripping voltammetry, ITO- Indium Tin Oxide, BieneNS@C: Bismuthene nanosheets with biomass-derived carbon, CPE: Carbon paste electrode, TRGOPE: Thermally reduced graphene oxide paste electrode, rGO-MWCNT/BDD: Reduced graphene oxide/multi-walled carbon nanotube on boron-doped diamond, E: Electrode, GE: Graphite electrode, PTH-afGQDs: poly(thionine)- amino-functionalized graphene quantum dots, Bi/A-GCE bismuth modified activated glassy carbon electrode, BC/Au: Biochar supported nano-gold, FTO: Fluorine-doped tin oxide, BC/Cu_2O: Porous activated carbon modified cuprous oxide

(MOF-808-His). Researchers created MOF-808 solvo-thermally and then post-synthesized it by exchanging ligands. This substitution increased the metal-binding affinity because histidine replaced coordinated acetate groups. Using square wave anodic stripping voltammetry (SWASV), the researchers detected Pb^{2+}, and the adjusted sensing parameters, which included the pH of the detection medium, the deposition period, and the deposition potential, resulted in outstanding analytical performance. The LOD, $S/N = 3$ of the sensors was exceptionally low at $(1.12 \pm 0.10) \times 10^{-10}$ M. The nanocomposite was successfully applied in tap water samples [20]. A new electrochemical sensor for the detection of Pb^{2+} ions in aqueous media was described by Li et al. in 2025. It is based on a composite material that consists of graphene integrated with metal-organic frameworks (NH_2-UiO-66) functionalized with ethylenediaminetetraacetic acid (EDTA). Heavy metal ion preconcentration was made possible by the NH_2-UiO-66 MOF's large specific surface area and distinct porous structure. The addition of graphene boosted the composite's electrical conductivity, accelerating electron transfer and enhancing signal sensitivity, while the grafted EDTA, which was abundant in –COOH groups, enabled robust chelation of Pb^{2+} and expanded the number of redox-active sites. With an outstanding detection limit of 0.052 μM, differential pulse anodic stripping voltammetry (DPASV) demonstrates a broad linear detection range of 0.1–50 μM for Pb^{2+} [79].

2.4.2 Use of graphene and graphene quantum-dot-based nanomaterials

In 2018 Balasubramanian et al. reported the voltametric detection of nitrobenzene (NB) from real water samples such as river, pond and sewage water using reduced graphene oxide (rGO) supported cellulose microfiber (CMF) nanocomposite. Regarding NB reduction, the CMF-rGO film modified glassy carbon electrode showed encouraging electrocatalytic activity in terms of high sensitivity 733.8 $\mu AmM^{-1}\ cm^{-2}$, wide linear range 0.2–927.7 μM, and low detection limit 88 nM in DPV. The nanocomposite

material was characterized by several analytical techniques such as Raman, FT-IR, SEM with EDX spectroscopy [29]. El-Badawy et al. In 2018, the researchers found that microwave irradiation for only 30 s created a quick and environmentally friendly technique for producing high-quality graphene nano sheets (GR) by simultaneously reducing and exfoliating graphite oxide. The total avoidance of dangerous reducing agents, intercalation chemicals, or chemical activation agents is a significant benefit of this strategy over previously reported techniques. TEM, FTIR, TGA, and XRD were utilized to confirm the formation of GR and its structural. With a low detection limit of 1.37 nM, the electrode demonstrated good analytical performance when used for the sensitive electrochemical detection of the antibacterial compound chloroxylenol. Since chlorinated phenolic chemicals are very harmful and persistent in the environment, the GR/CPE sensor was also used to find chloroxylenol in real samples, such as water and commercial formulations [80]. A thermally reduced graphene oxide paste electrode (TRGOPE) was reported in 2018 by Brycht et al. using thermally reduced graphene oxide (TRGO), which was prepared by thermally reducing graphene oxide in air. A variety of complementary techniques, such as FTIR, XRD, TEM, selected area electron diffraction (SAED), and atomic force microscopy (AFM), were used to characterize the TRGO's structure and morphology. The TRGOPE sensor was also used to detect the disinfectant and antiseptic 4-chloro-3,5-dimethylphenol (PCMX) with high sensitivity. By providing concentration throughout a linear dynamic range of 0.1–4.5 μM, by SWV response of PCMX with detection limit of 20.8 nM. TRGOPE's practicality was confirmed by analysing samples of tap and river water [81]. In 2018, Ahmad et al. reported the electrocatalyst based on graphene oxide (GO) sheets for the detection of nitrobenzene (NB). Several methods such as SWV, DPV, linear sweep voltammetry (LSV), and CV. The modified GO/GCE sensor showed a sensitivity of 157.14μA μM^{-1} cm^{-2}, 297.21μA μM^{-1} cm^{-2}, 677.14μA μM^{-1} cm^{-2}, and 1216.71 μA μM^{-1} cm^{-2}, as observed by CV, LSV, DPV, and SWV, respectively, and LOD of 66 nM by CV, 79 nM by LSV, 146 nM DPV, and 135 nM. Natural graphite flakes were used for the synthesis of GO and further characterized by several techniques such as FTIR, UV–vis, P-XRD, and FE-SEM spectroscopy [30].

In 2018, Castro et al. reported a highly sensitive electrochemical sensor based on an rGO/multi-walled carbon nanotube (rGO/MWCNT) nanocomposite for the detection of 2,4,6-trinitrotoluene (TNT) through square-wave voltametric (SWV). SWV shows a broad linear range 0.5–1100 μM with an LOD 0.019 μM for TNT. This novel substance holds considerable potential for the extremely sensitive detection of various analytes and other nitroaromatic explosives [32]. According to Yukird et al. (2018), because of their large surface area and semiconducting characteristics, zinc oxide (ZnO) nanorods have been thoroughly investigated for electrochemical sensing. After ZnO nanorods made by thermally breaking down zinc acetate, ZnO@graphene (ZnO@G) nanocomposites were recently prepared utilizing a simple room-temperature method and a colloidal coagulation effect. The ZnO nanorods, which had an average diameter of 89 ± 18 nm, were evenly distributed and intercalated between graphene layers, creating a stable nanocomposite, according to transmission electron microscopy. Square-wave anodic stripping voltammetry was used to optimize the sensor for the simultaneous electrochemical detection of Cd^{2+} and Pb^{2+} using the ZnO@G-modified electrode. The sensor showed dependable performance in real wastewater samples and, under ideal circumstances, exhibited a linear detection range of 10–200 μg L^{-1} with an LOD of 0.6 μg L^{-1} for Cd^{2+}

and 0.8 μg L^{-1} for Pb^{2+} [82]. Li et al. 2018, described the creation of an electrochemical sensor based on a nitrogen-doped carbon quantum dots and graphene oxide (NCQDs-GO) hybrid for the anodic stripping voltammetry-based simultaneous detection of Cd^{2+} and Pb^{2+} ions. Large electroactive surface areas and a wealth of oxygen-containing functional groups are provided by both NCQDs and GO in this hybrid system. These elements operate as active sites for the effective adsorption of heavy metal ions through electrostatic interactions, increasing sensor sensitivity. Because of their uniform distribution on the GO surface, the NCQDs make it easier for metals to re-oxidize during stripping analysis. With low detection limits of 7.45 μg/L for Cd^{2+} and 1.17 μg/L for Pb^{2+}, the NCQDs-GO-modified glassy carbon electrode (NCQDs-GO/GCE) demonstrated superior performance when compared to electrodes modified with NCQDs alone (NCQDs/GCE) or GO alone (GO/GCE). It achieved wide linear ranges of 11.24–11,241 μg/L for Cd^{2+} and 20.72–10,360 μg/L for Pb^{2+}. Additionally, the sensor showed high selectivity, repeatability, and adequate recoveries in actual water samples, such as tap and lake water [83]. In 2018 Ma et al. reported that highly sensitive electrochemical sensor based on ethanolamine (AE)- and silver nanoparticle (AgNP)-dual functionalized graphene oxide (Ag-AEfG) nanocomposites, which were created using a straightforward one-pot hydrothermal method. Several characterization methods was used to confirm the formation of nanocomposites such as XRD, FT-IR, SEM, TEM and XPS. In phosphate-buffered saline (PBS, pH 7.4), the Ag-AEfG-based sensor demonstrated remarkable nitrite detecting capability, with a detection limit of 0.023 μM ($S/N = 3$) and a wide linear range of 0.05–3000 μM. Additionally, the sensor showed outstanding sensitivity, stability, and repeatability. It was used successfully to detect nitrite in Nanjing tap water samples. AgNP–AEfG composites' potential for creating next-generation electrochemical sensing platforms is demonstrated by this work [35].

Sun et al. In 2018, Sun et al. reported using a simple electrochemical co-deposition technique to attach monodispersed metal nanoparticles (Cu, Pd, Pt, and Rh) evenly onto rGO nanosheets. SEM, or scanning electron microscopy, verified that the metal nanoparticles were evenly distributed across the rGO surface. Excellent electrocatalytic activity was demonstrated by the resultant nanocomposites in the electrochemical reduction of nitrate (NO_3^-) and nitrite (NO_2^-). Rh–rGO showed the maximum catalytic activity among them, and CV, XPS, and Fourier-transformed AC voltammetry were used to examine its reaction mechanism thoroughly. In a broad pH range (4–9), Rh–rGO demonstrated outstanding electrochemical stability, making it a viable electrocatalyst for the cleanup of nitrate-contaminated present in groundwater samples in the environment [36]. In 2018, Xuan et al. made a compact, pliable, and completely integrated electrochemical sensor that uses a flexible gold substrate with a micropatterned rGO-carbon nanotube (CNT) composite working electrode. Cadmium (Cd) and lead (Pb) ion anodic stripping peaks were made clear and distinct by the in-situ electrodeposition of a bismuth film. By increasing the electrode surface area, the integration of CNTs with rGO improved the sensor's overall performance and sensitivity. The sensor displays low detection limits of 0.6 ppb (Cd) and 0.2 ppb (Pb) under ideal circumstances. It was found that both ions had linear detection ranges of 20–200 ppb. The Au/rGO-CNT/Bi-modified electrode showed strong sensitivity, stability, and repeatability along with improved ion detection efficiency. Ultimately, the sensor effectively identified target heavy metal ions in drinking water samples [84] due to its good analytical performance.

In 2019, Ramachandran et al. reported excellent electrochemical sensing capabilities for nitroaromatic chemicals have been established by highly crystalline nitrogen-doped graphene quantum dots (N-GQDs), which are produced from polyaniline (PANI) using a straightforward hydrothermal process. High sensitivity and efficient distinction between different nitro compounds are demonstrated by the *N*-GQDs, underscoring their potential as selective sensors. For 2,4,6-trinitrophenol (TNP), the material's modified low LOD of 0.2 ppb (~ 1 nM) on a glassy carbon electrode (*N*-GQD/GCE) far exceeded previously documented micromolar-level sensitivity. The richly *N*-doped aromatic structure, which enhances conductivity and electron transfer while promoting selective interactions with nitroaromatics through π–π stacking, hydrogen bonding, or a combination of molecular interactions, is responsible for the superior sensing performance. The *N*-GQD/GCE is a viable platform for environmental monitoring of nitroaromatic contaminants because of these characteristics, which allow it to retain good sensitivity and selectivity in real water samples [85]. In 2019 Baghayeri et al. reported the poly(amidoamine) dendrimer functionalized magnetic graphene oxide (GO-Fe_3O_4-PAMAM) nanocomposite modified on glassy carbon electrode for the simultaneous detection of heavy metals ion such as Pb^{2+} and Cd^{2+} from real water samples through SWASV method. The modified sensor shows the wide linear range 0.4–120 and 0.2–140 μgL^{-1} for Pb2 + and Cd^{2+} respectively in SWASV. Additional metal ion potential interferents, such as Tl+, Cu^{2+}, Zn^{2+}, Hg^{2+}, Co^{2+}, Mg^{2+}, Ca^{2+}, Mn^{2+}, Ni^{2+}, Fe^{2+}, In^{2+}, and Cr^{2+}, were investigated but did not result in any notable interferences [86]. In 2020, Fu et al. reported that Nitrogen-doped graphene quantum dots (*N*-doped GQDs) have been successfully used to modify indium tin oxide (ITO) electrodes for the electrochemical detection of mercury ions (Hg^{2+}). *N*-doped GQDs were created by pyrolyzing citric acid and urea at 250 °C with the aid of infrared light. The average particle size of these GQDs was 4.5 nm. Electrochemical evaluation using EIS and CV showed that the equivalent series resistance and reductive and oxidative peak currents reduced consistently as the concentration of Hg^{2+} increased. *N*-doped GQD/ITO electrodes LOD of 10 ppb for Hg^{2+} and an accumulation period of 32 s. Additionally, the electrodes showed superior selectivity for Hg^{2+} in comparison to other metal ions. These findings highlight the potential of functionalized *N*-doped GQDs in environmental monitoring applications by showing that they offer a highly sensitive and selective platform for electrochemical detection of hazardous mercury ions in aqueous solutions [24].

In 2021 Kokulnathan et al. published report the detection of toxic nitroaromatic compound like nitrobenzene (NB) from real water samples through CV and DPV using magnesium oxide encapsulated on graphene oxide (GO/MgO) nanocomposite as sensor material. The sensor shows promotes quick electron transit to produce notable electrocatalytic activity and high sensitivity 11.6 μA μM^{-1} cm^2, spacious concentration range 0.1–38.9 and 58.5-2333.5 μM with LOD 0.01 μM at pH 7. The material was characterized through several technique XRD, Raman, SEM and EDX [31]. In 2021 Ruiz-Ramirez et al. reported the determination of nitrobenzene (NB) from synthetic water samples using CV method. Flower-like hierarchical rutile-phase TiO2 nanorod grown on graphene oxide sheet and its used sensor material for the detection of NB. This electrode material was synthesized via several analytical techniques such as XRD, TGA, Raman, SEM with EDX and TEM [87]. In 2021 Niu et al. reported the Si-doped graphene nanosheets (Si-GNS) as electrode material and modified onto GCE for the detection of

several nitroaromatic compounds such as nitrobenzene, 4-nitrotoluene, 2-nitrotoluene, 2, 4-dinitrotoluene and 2, 4, 6 trinitrotoluene through CV. Si-GNS was successfully prepared via thermal annelation method using sealed glass ampoule. Electrochemical catalytic mechanism was calculated using DFT method [88]. According to Saisree et al. (2021), the sensor potential of nitrogen-doped graphene quantum dots (*N*-GQDs) was recently investigated for the first time in relation to dangerous heavy metal ions, showing selective detection of cadmium ions (Cd^{2+}), a substance that is recognized by the us as a serious public health issue. The current response of aromatic *N*-GQDs, which were produced hydrothermally using polyaniline as a single precursor, was 25 times that of bare GCE and 6 times that of GQD-modified GCE (GQD/GCE), respectively. With very little current response from other interfering metal ions, the sensor demonstrated exceptional selectivity for Cd^{2+}. Using a pre-reduction procedure, the LOD was further increased to 1×10^{-7} ppb (parts per billion), which was the lowest recorded to date at 1×10^{-5} ppb. The sensitivity criteria for Cd^{2+} detection under the EPA acceptable limit of ≤ 5 ppb were met by the sensor, which showed a broad linear detection range (0.001–112,000 ppb). The sensor was also applied real water samples such as groundwater, seawater, and wastewater. Additionally, the sensor was used on real water samples, including wastewater, ocean, and groundwater [89]. According to Soman et al. (2021), a suspension polymerization approach was used to prepared a mercury ion-imprinted polymer (IIP) with a thiourea-derivatized graphene quantum dot (GQDTU) as the functional monomer. Using CV and DPV, the resultant GQDTU-IIP material was used to alter a GCE using a straightforward drop-casting technique. It was then assessed for Hg^{2+} ion detection. A modified electrode made of non-imprinted polymer (NIP) was also prepared for comparison. From 5×10^{-8} M to 2.3×10^{-5} M using DPV, the GQDTU-IIP-modified electrode had a good linear response to Hg^{2+} concentrations, with a detection limit of 23.5 nM. Two linear ranges were also identified by CV analysis, with a detection limit of 30.2 nM: 6×10^{-8} M to 8.5×10^{-7} M and 1.4×10^{-6} M to 7×10^{-6} M. The potential for environmental monitoring of trace mercury pollution was confirmed by the developed sensor's effective use in determining the levels of Hg^{2+} ions in real water samples and its dependable analytical performance [90]. In 2021, Hadidi et al. reported novel catalyst such as D-penicillamine functionalized graphene quantum dots (GQD-DPA) and validate through a variety of characterisation approaches, like FTIR, SEM and TEM etc. To create a Pb^{2+} electrochemical sensor, the GQD-DPA nanocomposite was used as an electrode modifier. The GQD-DPA-modified glassy carbon electrode (GQD-DPA/GCE), which has a larger surface area and several functional groups, showed noticeably improved sensitivity and selectivity toward Pb^{2+} detection. The sensor demonstrated a linear response with a detection limit of 0.01 nM across the Pb^{2+} linear concentration range of 0.03–2.8 nM. GQD-DPA/GCE also shown outstanding anti-interference capabilities and was effectively used to determine Pb^{2+} in tap and river water samples with high precision and accuracy [91]. In 2021, Sayyad et al. reported synthesizing and functionalizing a poly(3,4-ethylenedioxythiophene): poly(styrene sulfonate) and rGO nanocomposite (PEDOT: PSS/rGO) with L-cysteine (CySH). XRD, Raman spectroscopy, atomic force microscopy (AFM), and electrochemical studies were used to thoroughly evaluate the PEDOT: PSS/rGO/CySH nanocomposite. Excellent selectivity and high sensitivity toward Pb^{2+} detection were demonstrated by the electrochemical sensor. This sensor's linear concentration range is 1–70 ppb, and LOD is 0.09

ppb. This LOD exceeds previously reported electrochemical sensors and is significantly lower than the World Health Organization's (WHO) recommended maximum allowable Pb levels in drinking water (48 nM or 10 ppb) [92]. In 2023, J S et al. reported that partially rGO-molybdenum disulfide-modified electrodes (prGO-MoS_2/GCE) for selectively detection of Pb^{2+} and Cd^{2+} ions in real water samples. This Pb^{2+} and Cd^{2+} ranked among the top ten chemicals of primary public health concern by the World Health Organization (WHO) due to their widespread industrial use and severe health effects. Notably, the sensor displayed negligible response to other metal ions (Hg^{2+}, Ag^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+}, Zn^{2+}, Sn^{2+}, Cu^{2+}, and Cr^{4+}), demonstrating excellent selectivity. The LOD were extremely low ~ 0.0002 ppt for Pb^{2+} and 0.1 ppt for Cd^{2+} -well below the WHO safe limits of ≤ 10 ppb and ≤ 5 ppb, respectively. The superior sensing performance is attributed to the synergistic structure of prGO–MoS_2, where strong bonding of MoS_2 to prGO enhances electron transfer and metal ion affinity. These findings underscore the potential of prGO-MoS_2-based electrodes for highly sensitive and selective monitoring of toxic heavy metals in drinking water [93]. In 2023, Saisree et al. By employing sulfuric acid as an acid catalyst and S-doping agent in a straightforward hydrothermal process, a sulfur-and-nitrogen-co-doped graphene quantum dot (S, N-GQD) was synthesised from polyaniline, yielding highly crystalline, aromatic nanosized quantum dots (~ 4 nm). With strong, well-resolved peaks and sufficient peak-to-peak separation, this material demonstrated simultaneous electrochemical sensing of three of the top 10 hazardous heavy metals listed by the WHO: Cd^{2+}, Pb^{2+}, and Hg^{2+}. Comparatively, the Cd^{2+} signal on S, N-GQDs was almost seven times greater than that on N-GQDs, while control N-GQDs without S-doping only displayed current response for Cd^{2+}. The LOD for simultaneous sensing were very low: 1 pM for Hg^{2+}, 10 pM for Pb^{2+}, and 1 pM for Cd^{2+}. The sensor demonstrated its practical usefulness for environmental monitoring by maintaining its performance in genuine environmental water samples, such as groundwater, saltwater, and wastewater spiked with the metal ions. It achieved about 100% recovery with a relative standard deviation of less than 0.50% [94]. In 2023, Rana et al. published that a simple one-step hydrothermal method was used to successfully prepared a two-dimensional (2D) layered nanocomposite made of rGO and molybdenum disulfide (MoS_2). Vertically oriented MoS_2 nanosheets were consistently grown on a horizontally aligned rGO framework, creating a well-ordered layered architecture, according to structural and morphological characterisation. Further characterised by XRD, Raman, and XPS. The MoS_2-rGO nanocomposite's electrochemical performance was systematically assessed for mercury ion (Hg^{2+}) sensing; cyclic voltammetry studies showed a ~ 103% increase in oxidation peak current response at the MoS_2-rGO-modified electrode compared to a bare electrode, highlighting its superior electrochemical activity; the sensor showed excellent analytical performance, achieving a LOD of 1.6 μM, a sensitivity of 76.302 μM μA^{-1} cm^{-2}, and a limit of quantification (LOQ) of 5.4 μM. These results show that the MoS_2-rGOnanocomposite is a promising candidate for the development of next-generation electrochemical sensors for heavy metal ion detection, offering excellent sensitivity, stability, and scalability for practical real water applications [95].

In order to detect mercury ions in a water sample, Tian et al. (2020) reported using a one-pot electrochemical technique based on cyclic voltammetric scanning to create a GCE modified with a nanocomposite made of poly(thionine) and amino-functionalized graphene quantum dots (afGQDs). With a detection limit of 0.6 pM, the sensor

has an excellent linear range from 1 pM to 1 μM at pH 6.5. For real water samples, a good recovery rate was also noted, ranging from 97.00 to 103.00%. Nanomaterial was characterised by several analytical techniques such as UV–visible, FTIR and SEM spectroscopies [70]. According to Visagamani et al. (2023), a new $SrTiO_3$/Ag/rGO composite was synthesized for electrochemical sensing applications using an environmentally friendly ultrasonic-assisted technique. The $SrTiO_3$/Ag/rGO/SPCE sensor was created by integrating the composite onto a screen-printed carbon electrode (SPCE) to detect 4-NP. Using Raman spectroscopy, field-emission SEM, FTIR, XRD, and UV-visible spectroscopy, the material was methodically characterized. EIS verified the $SrTiO_3$/Ag/rGO composite's increased electrical conductivity. In comparison to previously published modified electrodes for 4-NP measurement, LSV demonstrated an ultralow LOD (0.03 μM) and a wide linear detection range (0.1–1000 μM). With recovery rates ranging from 97.50 to 98.70% in river and drinking water samples, the $SrTiO_3$/Ag/rGO/SPCE sensor also demonstrated outstanding selectivity, stability, and repeatability [71]. Highly sensitive electrochemical sensor for 4-NP detection was reported by Patel et al.. in 2025. It was based on a novel nanocomposite made of rGO that was bioinspired, phyto-synthesized, and grafted with silver nanoparticles and graphitic carbon nitride (AgNPs–rGO). The material was made using a one-pot green method in which graphene oxide (GO) and silver nitrate were both reduced simultaneously using Ocimum sanctum leaf extract as a natural reducing agent. This method provided a sustainable and economical synthesis method. The prepared sensor showed an extraordinarily broad linear dynamic range, covering 0.015–846 μM and 846–8001 μM, as well as an ultralow detection limit of 15 nM. Additionally, its practical practicality, high sensitivity, and repeatability were demonstrated in a real water sample analysis, thus validating its application [72]. According to a 2025 study by Dhrishya et al. heavy metal ions (HMIs) including Pb^{2+}, Hg^{2+}, Cd^{2+}, and As^{3+} have become a significant environmental problem because of their extreme toxicity, even at ppb (parts-per-billion) levels. HMIs in environmental water sources must thus be continuously monitored. The electrochemical detection of Pb^{2+} ions was recently shown using a hybrid quantum dot material made of boron nitride (BN) and nitrogen-doped graphene (Hyb-BNQD-NGQD), which was produced from bulk BN and polyaniline (PANI). The Hyb-BNQD-NGQD demonstrated an exceptionally high current responsiveness and an ultra-low detection limit of 1×10^{-12} M (0.2 ppt) in contrast to unmodified glassy carbon electrodes, BNQDs, NGQDs, and physical combinations of BNQDs and NGQDs. The material demonstrated consistent performance in wastewater samples (≤ 5% variance) and maintained great selectivity for Pb^{2+} even in the presence of other potentially interfering metal ions, demonstrating its outstanding specificity and usefulness for heavy metal monitoring [96].

2.4.3 Use of activated carbon/biochar-based nanomaterials

In 2019 Zhang et al. published an activated carbon (PAC) with homogeneous decoration and palladium nanoparticles (Pd NPs) modified glassy carbon electrode (Pd@PAC/GCE) based sensor for the detection of Cd^{2+}, Pb^{2+} and Cu^{2+}in real samples. The SWASV technique was applied for sensing and they found LOD 13.33, 6.60 and 11.92 nM for Cd^{2+}, Pb^{2+} and Cu^{2+} respectively. The composite material was characterized by several analytical techniques XRD, Raman, SEM and TEM [97]. In 2022, Harraz et al. revealed amperometric detection of 4-NP utilizing a sensitive and selective electrochemical sensor based

on a new gold nanoparticle/rGO/activated carbon nanocomposite. The activated carbon was prepared from dates seeds through pyrolysis and further combined with on AuNPs/rGO. The sensor was able to detect the 4-NP using the amperometric approach with a high sensitivity of 0.03436 $\mu A \mu M^{-1}$ throughout a broad linear dynamic range (LDR) of 18–592 μM and a LOD of 0.36 μM at (S/*N* = 3) [28]. In 2023 Aravind et al. reported the Activated Carbon- reduced Graphene Oxide (AC-rGO) modified carbon paste electrode (CPE) based electrochemical sensor for simultaneous detection of Pb^{2+} and Cd^{2+} in wastewater samples. CV and LSV was used as sensitive technique for the detection of Pb^{2+} and Cd^{2+} with LOD 10.91 μg L^{-1} and Pb^{2+} as low as 14.01 μg L^{-1} respectively. The activated carbon was prepared from rubberwood saw dust and further characterized by FT-IR and BET surface area [21]. In 2023 Beas-Bernuy et al. reported a novel electrochemical sensor based on orange peel activated carbon (ACOP) derived from biomass and multiwalled carbon nanotubes (MWCNTs) composite coated on CPE for detection of Cd^{2+} in real samples. The CV and SWV was applied for the detection of Cd^{2+} with a limit of quantification of 3.00 ± 2.60 μM, a LOD of 0.91 ± 0.79 μM, and a coefficient of determination r^2 of 0.9907. The nanocomposite materials were characterized by FTIR, Raman, and EIS [98].

According to Yu et al. (2023), bismuth dendrites were electrodeposited onto an activated GCE in order to detect 4-NP, a hazardous aromatic water contaminant, electrochemically. Bismuth dendrites were electrode deposited at − 1.0 V vs. SCE for 400 s after potential cycling between − 2.0 and + 2.0 V vs. SCE activated the GCE by creating active sites that promoted electron transport. The sensor demonstrated two linear detection ranges (0.005-1.6 μM and 1.6–170 μM), with a sensitivity of 29.4 μA μM^{-1} cm^{-2} for the lower concentration range and a detection limit of 0.18 nM. The electrode's recovery rates in spiked water samples ranged from 96.50 to 106.70%, demonstrating exceptional repeatability, reproducibility, and selectivity [73]. Zhang et al. proposed a new electrochemical sensor in 2023 that uses gold nanoparticles (AuNPs) adsorbed on charcoal (BC) as the functional sensing material to detect nitrite with high sensitivity and selectivity. XRD, FT-IR, Raman, and SEM were used to extensively analyze the BC/Au composite, which was created using a straightforward and economical process. The linear range of this sensor is 0.5 μM to 6 mM, and its detection limit is 0.14 μM. This sensor was also used to analysed in real samples [34]. The ultra-trace detection of Hg^{2+} in aquatic settings was reported by Zou et al. in 2023. Through a straightforward impregnation-pyrolysis process, porous activated carbon (BC) was modified with cuprous oxide (Cu_2O) to prepared a BC/Cu_2O composite, which was then used to create a novel electrochemical sensor. DPASV shows broad linear detection range of 1.0 ng L^{-1} to 1.0 mg L^{-1} and a low detection limit of 0.3 ng L^{-1}. The sensor demonstrated good sensitivity. These findings demonstrate the BC/Cu_2O composite's potential as a sensitive and trustworthy electrochemical platform for Hg^{2+} detection in environmental samples [25]. A simple synthesis of bismuthene nanosheets (BieneNS) and its incorporation with carbon generated from biomass to create BieneNS@C composites were described by Zhang et al. in 2025. TEM and atomic force microscopy (AFM) showed the layered hexagonal structure of BieneNS, while XRD and XPS proved the successful synthesis and chemical composition of BieneNS@C. Excellent conductivity and strong current responses toward Cd^{2+} detection were shown by electrochemical characterisation using CV, DPV, and EIS. GCE were modified using the BieneNS@C composites, resulting in a highly

sensitive electrochemical platform for the detection of Cd^{2+} in aqueous solutions. The sensor demonstrated a LOD of 0.2 ppb and a broad linear response range of 0–150 ppb under ideal circumstances. The nanocomposite was successfully used for Cd^{2+} trace-level detection in intricate water systems and environmental monitoring [99].

3 Future perspectives and research directions

Electroactive advanced nanomaterials, such as MOF-based nanocomposite, graphene and graphene quantum dots-based nanocomposite, and activated carbon and biochar-based nanocomposite, have become a game-changer for wastewater treatment because of their high surface area, adjustable electronic characteristics, and improved catalytic efficiency. Future studies should concentrate on the logical creation of nanomaterials with ideal electrochemical characteristics to allow for the effective and selective removal of a variety of pollutants, such as nitrate contaminants, heavy metals ions, and nitroaromatic compounds. Charge transfer, conductivity, and catalytic stability can be further enhanced under real wastewater circumstances by merging several functional components, creating a composite with another metal, or engineering hierarchical nanostructures.

The development of economical and environmentally friendly electroactive materials is essential to reduce costs and minimize environmental impact in the monitoring of toxic pollutants present in wastewater. To address this need, we also focus on bio-derived materials, such as biochar and activated carbon-based materials, as promising electroactive and sustainable nanomaterials for future applications.

4 Conclusion

In summary, this review highlights recent advances (2018–2025) in the development of electroactive nanomaterials for environmental water monitoring and treatment. Advanced electroactive nanomaterials have become essential to the creation of sustainable nanomaterials for monitoring and remediation of toxic pollutants. MOFs, graphene-based nanostructures, and materials produced from activated carbon are among those that have shown remarkable electrochemical performance, adjustable physicochemical characteristics, and versatility for a range of environmental applications. MOFs provide unmatched porosity and structural plasticity, enabling tailored pollutant collection and functionalization, while graphene-based materials contribute excellent electrical conductivity and large surface area, facilitating quick electron transfer and better sensing performance. Nanomaterials based on activated carbon and biochar, which are produced from plentiful and renewable resources, also provide a low-cost, eco-friendly and sustainable materials substitute with remarkable stability and adsorption capability. Heavy metal ions, nitroaromatic compounds, nitrate, and haloaromatic pollutants may all be detected in real time using these materials because of their distinctive electrochemical characteristics, large surface area, adjustable porosity, and chemical adaptability.

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Author contributions

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Clinical trial number
Not applicable.

Ethics approval and consent to participate
Not applicable.

Consent to participate
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Competing interests
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